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PARAMPARA 1.0

A inter college Hackathon of Ramakrishna Mission Vivekananda
Centenary College

Organized By: E-cell RKMVCC

Event Timeline

Aug 10 – Registration Opens

Aug 20 – Registration Closes

Aug 21 – Final Problem Statement Submission Deadline

Aug 22 – Pitching Presentation & Final Evaluation Day

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Problem Statements & Domain Categories

0.1 Overview of Allowed Themes

Participating teams at RKMVCC Rahara must choose a problem statement categorized under one of the 8 approved themes listed below:

1. Agriculture, FoodTech & Rural Development
2. Clean & Green Technology
3. Tourism
4. Blockchain & Cybersecurity
5. Smart Education
6. Disaster Management
7. MedTech / BioTech / HealthTech
8. Miscellaneous (Open Innovation)

0.2 Specific Problem Statements (PS Catalog)

Teams can select any one of the following pre-defined Problem Statements for their hackathon entry:

0.2.1 Theme 1: Agriculture, FoodTech & Rural Development

- **PS-AFR-01: AI-Driven Multi-Modal Satellite Analytics for Crop Mapping, Moisture Stress, and Irrigation Advisories**
 - **Track / Project Type:** Software Track (Geospatial AI / Machine Learning Engine & Dashboard)
 - **Problem Description:** Agricultural monitoring across large command areas is heavily impeded during cloud-heavy monsoon seasons, making traditional single-source optical satellite data unreliable. Furthermore, existing systems treat crop classification and moisture stress as isolated metrics without accounting for stage-wise crop phenology or translating water deficits into actionable field-level irrigation guidance. The lack of integrated, all-weather, near-real-time satellite analytics leads to avoidable irrigation losses, delayed drought detection, and unverified crop damage assessments in government schemes.
- **PS-AFR-02: Micro-Cold Storage Monitoring & IoT-Based Post-Harvest Logistics Network**
 - **Track / Project Type:** Hardware Edition Track (IoT Sensors + Embedded Firmware + Cloud Dashboard)

- **Problem Description:** India experiences significant post-harvest losses in fruits, vegetables, and dairy due to an inadequate cold chain infrastructure during the “first-mile” transit from farm gates to regional markets. Small farmers cannot afford large-scale refrigeration, while existing micro-cold rooms and solar-powered cooling units often suffer from unmonitored temperature/humidity fluctuations, power cutouts, and inefficient logistics scheduling, leading to premature food spoilage.
- **PS-AFR-03: Optimizing Urban Heat Mitigation and cooling strategies via Artificial Intelligence and Machine Learning (AIML)**
 - **Track / Project Type:** Software Track (Geospatial AI / Machine Learning Engine & Dashboard)
 - **Problem Description:** Develop a geospatial AI/ML-based system, backed with physics informed decision making to identify urban heat stress hotspots, quantify key drivers of urban heating, and generate optimized, scenario-based cooling interventions for mitigating urban heat impacts.

0.2.2 Theme 2: Clean & Green Technology

- **PS-CGT-01: AI-Driven Building Energy & Carbon Audit Engine for Educational Campuses**
 - **Track / Project Type:** Software Track (Data Analytics / IoT Integration / Web Dashboard)
 - **Problem Description:** Educational institutions and commercial facilities consume massive amounts of energy through inefficient HVAC systems, unmonitored lighting, and idle electronic equipment left running after hours. Institutional managers lack granular visibility into building-by-building or floor-by-floor power consumption, making it difficult to spot energy leaks, track carbon footprints, or meet government sustainability compliance benchmarks.

0.2.3 Theme 3: Tourism

- **PS-TR-01: Authentic Tribal Homestay Aggregator & Cultural Eco-Tourism Marketplace**
 - **Track / Project Type:** Software Track
 - **Target Audience:** Rural Indigenous Communities, Tribal Homestay Hosts, Local Artisans, and Eco-Conscious Tourists
 - **Problem Description:** While North East India possesses immense potential for eco-tourism and cultural immersion (e.g., ethnic village homestays, indigenous handicraft workshops, local tribal festivals), mainstream commercial OTA (Online Travel Agency) platforms fail to reach rural home-hosts due to digital literacy gaps, language barriers, and formal banking requirements. Consequently, major tourist spending remains concentrated among urban operators, while local indigenous communities miss out on economic opportunities.

0.2.4 Theme 4: Blockchain & Cybersecurity

- **PS-BC-01: Secure Encrypted Closed-Group Communication Platform for Defence Personnel over Public Mobile Networks**
 - **Track / Project Type:** Software Track (Android/iOS Mobile Client + Encrypted Server Backend)
 - **Problem Description:** Military personnel and their families routinely communicate using commercial messaging platforms over public cellular and Wi-Fi networks. Commercial communication channels pose severe operational security (OPSEC) risks due to data harvesting, server vulnerabilities, location tracking, and lack of sovereign cryptographic control. Commercial platforms are prone to interception or targeted exploitation by adversary signal intelligence agencies.
- **PS-BC-02: Blockchain-Based Immutable Provenance & Tracking System for Ordnance Ammunition & Military Supply Chains**
 - **Track / Project Type:** Hybrid (Blockchain Distributed Ledger + Smart Contracts + Serialization Hardware)
 - **Problem Description:** Managing military logistics—specifically ammunition, missile components, weapons, and sensitive spare parts—requires absolute transparency, tamper-proof audit trails, and strict origin verification. Traditional centralized inventory databases in defence supply chains are susceptible to unauthorized database tampering, unauthorized batch swaps, counterfeit component insertion, and poor visibility during inter-depot transit in remote border sectors.

0.2.5 Theme 5: Smart Education

- **PS-ED-01: Secure & Tamper-Proof Question Paper Leakage Prevention System**
 - **Track / Project Type:** Hybrid (Hardware + Software) or Software-Only
 - **Problem Description:** Question paper leaks before or during high-stakes competitive and university examinations undermine educational integrity, cause massive financial losses, and distress millions of students. Traditional distribution models rely heavily on physical logistics, manual handling, and static storage, creating multiple vulnerable attack vectors at printing presses, transit hubs, and institutional vaults. Existing digital methods often lack end-to-end cryptographic protection or dynamic access controls right up to the minute the exam begins.
- **PS-ED-02: Algorithmic Skill-Matching & Team Formation Platform for Students**
 - **Track / Project Type:** Software Track (Web / Mobile Application)
 - **Problem Description:** In higher education institutions, students frequently struggle to form balanced, multidisciplinary teams for hackathons, research initiatives, and technical projects. Isolated developers, designers, and domain

strategists often lack visibility into the broader campus talent pool. Relying on informal word-of-mouth or unorganized messaging channels results in skill mismatches, incomplete project rosters, and underutilized student potential.

0.2.6 Theme 6: Disaster Management

- **PS-DM-01: AI-Powered Digital Twin of India's Climate using India's National Data**

- **Track / Project Type:** Software Track (AI/ML Models + High-Performance Computing Pipeline + Geospatial Visualization Dashboard)
- **Problem Description:** Traditional climate models in India often lack the spatial and temporal resolution required for effective localized adaptation strategies against climate change. Furthermore, these conventional models are computationally intensive, hindering near-real-time simulations. There is a critical national need for a high-fidelity, dynamic virtual replica (Digital Twin) of India's climate system that continuously evolves. This twin must integrate diverse, indigenous observations (from Indian satellites like INSAT/Oceansat and ground-based IMD networks) and leverage advanced AI to simulate atmospheric, oceanic, and land-surface processes at high resolution. Such a system is essential to accurately capture complex local phenomena like monsoon variability, extreme precipitation events, and drought evolution with greater accuracy than current standard models.

0.2.7 Theme 7: MedTech / BioTech / HealthTech

- **PS-MBH-01: Digital Traceability & Botanical Authentication Platform for Ayurvedic Medicinal Herbs**

- **Track / Project Type:** Software Track (Web Application / Database Management System)
- **Problem Description:** Adulteration and accidental misidentification of medicinal plants (e.g., substituting *Bacopa monnieri* with look-alike species) pose severe risks to herbal drug efficacy and consumer safety. Raw herbal supply chains in India are fragmented, making it difficult for pharmaceutical buyers to verify whether harvested plants possess the correct phytochemical composition, geographic origin, or morphological characteristics required for official Ayurvedic pharmacopoeias.

- **PS-MBH-02: AI-Powered Antimicrobial Resistance (AMR) Tracker & Microbial Susceptibility Predictor**

- **Track / Project Type:** Software Track (Web Dashboard / Machine Learning Analytics Engine)
- **Problem Description:** Antimicrobial Resistance (AMR) is a growing global health crisis. In clinical settings, waiting for traditional disk-diffusion culture results (24–48 hours) often forces doctors to prescribe broad-spectrum antibiotics empirically, accelerating bacterial resistance. Furthermore, diagnostic laboratories lack unified software tools to analyze local minimum inhibitory

concentration (MIC) trends across bacterial strains (*E. coli*, *S. aureus*, *K. pneumoniae*) to guide targeted antibiotic stewardship.

- **PS-MBH-03: Digital Vector Ecology & Zoonotic Disease Early-Warning Dashboard**

- **Track / Project Type:** Software Track (GIS Mapping / Predictive Analytics Web Portal)
- **Problem Description:** Outbreaks of vector-borne and zoonotic diseases (e.g., Dengue, Malaria, Chikungunya, Scrub Typhus, and Leptospirosis) are closely tied to the biological life cycles of vectors like *Aedes/Anopheles* mosquitoes, ticks, and rodent hosts. Current municipal control measures are largely reactive—spraying insecticides only after human infection clusters are reported. There is a lack of predictive software integrating vector breeding biology, larval density metrics, host mammal populations, and ambient weather parameters to intervene before outbreaks occur.

0.2.8 Theme 8: Miscellaneous (Open Innovation)

- **PS-MS-01: Open Societal Innovation Challenge** — Teams may define their own problem statement addressing a pressing real-world issue in campus life, civic administration, or local community, provided it does not strictly fit the above 7 themes.